

[SCHOOL NAME — PLACEHOLDER LETTERHEAD]

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Class 4 Maths — CA-1 (Medium) — Answer Key

Class : 4

Subject : Maths (Math-Mela)

Max Marks : 70

Duration : 2 Hr 30 Min

Syllabus: Chapters 1–3 — Shapes Around Us, Hide and Seek, Pattern Around Us

General Instructions: All questions are compulsory. For Section A, write the correct option letter.

ANSWER KEY

SECTION A — Multiple Choice Questions (1 Mark each)

1. A cuboid has 6 faces, 12 edges, and how many corners, following Faces + Corners - Edges = 2? (1 Mark)

A. 6

B. 8 ✓

C. 10

D. 12

2. Which of these shapes has NO straight sides or angles at all? (1 Mark)

A. Triangle

B. Square

C. Circle ✓

D. Rectangle

3. If a shape has 6 sides, how many angles does it have? (1 Mark)

A. 4

B. 5

C. 6 ✓

D. 8

4. If Mini is looking at a building from a helicopter directly above it, which view is she seeing? (1 Mark)

A. Front view

B. Side view

C. Top view ✓

D. None of these

5. On a grid, an object at row 3, column 1 is located _____. (1 Mark)

A. in the top-right corner

B. in the third row, first column ✓

C. outside the grid

D. in the first row, third column

6. Which statement about odd and even numbers is TRUE? (1 Mark)

A. An odd number plus an odd number is always odd

B. An even number plus an even number is always even ✓

C. All numbers are even

D. Odd numbers cannot be added

7. To pay exactly ₹75 using the fewest notes/coins, which combination works best (using 50, 20, 5)? (1 Mark)

A. 50 + 20 + 5 ✓

B. 20 + 20 + 20 + 5 + 5 + 5

C. 5 × 15

D. 50 + 5 × 5

8. A prism and a pyramid both have flat faces, but which of these is a key difference between them? (1 Mark)

A. A prism has two identical matching bases; a pyramid has only one base ✓

B. A pyramid has more faces than any prism

C. Prisms cannot have rectangular faces

D. There is no difference between them

9. If a coin collection has 24 coins, can they always be grouped into complete pairs? (1 Mark)

A. Yes, because 24 is even ✓

B. No, because 24 is odd

C. It depends on the coin size

D. Coins cannot be paired

SECTION B — Short Answer Questions (2 Marks each)

1. Explain how you would identify whether a 2D shape is a pentagon or a hexagon. (0 Marks)

► You count the number of sides (or angles) - a pentagon has 5 sides and 5 angles, while a hexagon has 6 sides and 6 angles, so counting sides helps tell them apart.

2. A cube and a cuboid both have 6 faces, 12 edges, and 8 corners. Explain how you would tell them apart just by looking. (0 Marks)

► A cube has all six faces as equal squares, while a cuboid has rectangular faces that are not all the same size, so checking if all faces look identical helps tell them apart.

3. Explain why the top view of a round object like a ball looks different from the top view of a cube. (0 Marks)

► The top view of a ball (sphere) would show a circle, since it looks round from every angle, while the top view of a cube would show a square, since a cube has flat square faces.

4. If you are standing at the school gate and a friend is on the rooftop looking down, explain why you would see two different views of the same building. (0 Marks)
- You would see the front view (height, doors, windows) from the gate, while your friend would see the top view (the outline/layout of the roof and surroundings) from above, because each position offers a different angle of observation.
5. Explain how a grid can help someone find a hidden object in a "treasure hunt" game. (0 Marks)
- A grid divides an area into labelled rows and columns, so clues like "row 3, column 4" can point to an exact location, making it easier to search systematically rather than randomly.
6. A group of 33 students wants to form pairs for a game. Explain what will happen and why. (0 Marks)
- Since 33 is an odd number, the students cannot form complete pairs - 16 pairs will be formed, and one student will be left without a partner.
7. Explain two different ways to make a total of ₹60 using notes/coins of value 50, 20, 10, and 5. (0 Marks)
- One way: a ₹50 note and a ₹10 note ($50 + 10 = 60$). Another way: three ₹20 notes ($20 + 20 + 20 = 60$).
8. Why might builders use knowledge of prisms and pyramids when designing real buildings or monuments? (0 Marks)
- Understanding the properties of prisms (like stability from flat, matching bases) and pyramids (like strength from a pointed, tapering shape) helps builders choose the right structure for stability, appearance, and purpose.
9. Explain the pattern you would notice if you listed all odd numbers from 1 to 15. (0 Marks)
- The pattern is 1, 3, 5, 7, 9, 11, 13, 15 - each number increases by 2 from the previous one, showing a consistent, repeating pattern of adding 2 each time.
10. If a 2D shape has exactly 4 right angles and all four sides equal, what shape is it, and how do you know? (0 Marks)
- It is a square, because a square is defined by having 4 equal sides and 4 right angles (90 degrees each).
11. Explain why a map of a classroom (top view) would not show the height of the furniture. (0 Marks)
- A top view only shows what is visible from directly above, such as the outline and position of furniture, but not their height, since height can only be seen from a front or side view.
12. A vendor sells items and always gives change using the fewest coins/notes possible. Explain why this is a practical strategy. (0 Marks)
- Using the fewest coins/notes makes transactions quicker and reduces the amount of change both the vendor and customer need to carry or count, making the process more efficient.
13. Explain the difference between an edge and a face using a real object like a matchbox. (0 Marks)
- A face of the matchbox is one of its flat rectangular surfaces, while an edge is the straight line where two of these faces meet, such as where the top surface meets a side surface.
14. Why might understanding grid positions be useful when playing a strategy game like chess or Battleship? (0 Marks)
- These games use grids with labelled rows and columns to describe exact positions of pieces or ships, so understanding grid positions helps players communicate moves and locations accurately.

SECTION C — Short Answer Questions (Explain) (3 Marks each)

1. Explain in detail how you would determine the number of faces, edges, and corners of a triangular prism, and describe what makes it different from a triangular pyramid. (0 Marks)
- A triangular prism has 5 faces (2 triangular bases and 3 rectangular side faces), 9 edges, and 6 corners.
 - A triangular pyramid has 4 faces (1 triangular base and 3 triangular side faces meeting at a point), 6 edges, and 4 corners.
 - The key difference is that a prism has two matching, identical bases connected by rectangular faces, while a pyramid has only one base with triangular faces meeting at a single top point (apex).
2. Describe a real-world scenario where understanding top, front, and side views would be genuinely useful, and explain why. (0 Marks)
- An architect designing a house needs to show top, front, and side views (called elevations) so builders understand the full structure from every angle.
 - The top view shows the layout of rooms, the front view shows the height and appearance of the entrance, and the side view shows the depth of the house.
 - Without all three views, builders might misunderstand the design, leading to construction mistakes, showing why multiple views are genuinely useful in real life.
3. Explain how patterns in even and odd numbers can help predict outcomes, using the example of pairing students for a group activity. (0 Marks)
- If the total number of students is even, they can always be split into complete pairs with no one left out.
 - If the total is odd, one student will always be left without a pair, no matter how the students are arranged.
 - This shows that simply knowing whether a total number is even or odd lets us predict the outcome of pairing activities before actually trying to pair anyone.
4. A shopkeeper needs to give ₹235 in change using the fewest possible notes and coins (available: ₹200, ₹100, ₹50, ₹20, ₹10, ₹5). Explain the best combination and your reasoning. (0 Marks)
- To minimise the number of notes/coins, start with the largest value that fits: one ₹200 note covers most of the amount, leaving ₹35.
 - For the remaining ₹35, use one ₹20 note and one ₹10 note and one ₹5 coin ($20 + 10 + 5 = 35$).
 - Total combination: $₹200 + ₹20 + ₹10 + ₹5 = ₹235$, using only 4 notes/coins, which is the most efficient combination.
5. Explain how the relationship between the number of sides and angles in 2D shapes can help you identify an unknown shape quickly. (0 Marks)
- Since the number of angles in any 2D shape always equals its number of sides, counting either one gives you the same useful information.
 - For example, if you count 5 angles in a shape, you immediately know it has 5 sides, meaning it is a pentagon, without needing to count the sides separately.
 - This relationship makes shape identification faster and helps avoid mistakes when only part of a shape is visible or clear.

6. Discuss why both 2D shape knowledge (Shapes Around Us) and spatial view knowledge (Hide and Seek) are useful together when reading a building blueprint. (0 Marks)

- Blueprints use 2D shapes to represent rooms, walls, and furniture layouts as seen from above (a top view).
- Understanding different views (top, front, side) helps someone reading the blueprint imagine the full 3D structure of the building, not just a flat drawing.
- Combining both skills - recognising 2D shapes and understanding views - allows someone to accurately interpret a blueprint and picture the real building it represents.

SECTION D — Long Answer Questions (5 Marks each)

1. Design a simple 3D model (like a small house or toy) using at least three different 3D shapes learnt in "Shapes Around Us." Describe each shape used, its properties (faces, edges, corners), and why you chose it for that part of your model. (0 Marks)

- For the main body of the house, I would use a cuboid, which has 6 rectangular faces, 12 edges, and 8 corners, providing a stable, box-like structure for walls.
- For the roof, I would use a triangular prism or pyramid shape, which has sloped triangular faces that allow rain and debris to slide off easily.
- For a decorative element like a chimney, I could use a cylinder, which has a curved surface and two flat circular faces, giving a rounded look that contrasts with the cuboid body.
- Each shape was chosen based on its properties - stability from the cuboid, a sloped design from the prism/pyramid, and a rounded contrast from the cylinder - showing how understanding shape properties helps in practical design.

2. Explain in detail how a drone flying over your school, combined with a grid map of the classroom, could be used together to create a complete "guide" to your school for a new student. Describe what information each part (drone view and grid map) would provide. (0 Marks)

- A drone flying above the school would provide a top view, showing the overall layout - where the playground, main building, and pathways are located relative to each other.
- A grid map of a classroom would provide detailed positions within a smaller space, such as where each student's desk or important classroom items (like the teacher's table or the bookshelf) are located, labelled by row and column.
- Together, the drone's top view would help a new student understand the overall school layout, while the classroom grid map would help them find specific, detailed locations within a room.
- This combination shows how different scales of spatial understanding - a wide overview and a detailed close-up - work together to create a genuinely useful guide for someone unfamiliar with the school.

3. Explain, using detailed examples, how patterns (even/odd numbers, money combinations, and repeating sequences) appear in different parts of daily life, and discuss why recognising patterns is an important mathematical skill. (0 Marks)

- Even and odd number patterns appear when grouping objects, such as arranging chairs in pairs for an event, where knowing if the total is even or odd helps plan seating without leftover chairs.
- Money patterns appear when making payments, since understanding common note/coin values (like 5, 10, 20, 50, 100) helps people quickly figure out efficient ways to pay or give change.
- Repeating sequences appear in everyday designs, like patterns on clothes, floor tiles, or even calendars (where days of the week repeat every 7 days).
- Recognising these patterns is an important mathematical skill because it helps us predict outcomes, solve problems more efficiently, and understand the underlying order in the world around us, which is a foundation for more advanced maths later on.